

Private aur Protected Properties/Methods

Idea: encapsulation

Coffee machine ko chalane ke liye sirf button dabana aana chahiye, andar ke tar-pipe nahi. Object me bhi **internal interface** (andar ka kaam) ko **external interface** (bahar se use hone wala) se alag rakhte hain.

Fayde:

- User galti se andar ki cheezein tod nahi paata.
- Class likhne wala internals ko freely badal sakta hai, bahar ka code nahi tootta.
- Complexity chhup jaati hai, use karna aasan ho jaata hai.

Teen type ke fields

Type	Kaun access kar sakta hai	JS me
Public	sab	normal
Protected	class aur uski child classes	sirf convention (<code>_naam</code>)
Private	sirf wahi class	language level (<code>#naam</code>)

Protected: `_` prefix

Ye sirf ek convention hai, JS rokta nahi. "Bahar se mat chhuo."

```
javascript

class CoffeeMachine {
  _waterAmount = 0;

  set waterAmount(value) {
    if (value < 0) value = 0;
    this._waterAmount = value;
  }

  get waterAmount() {
    return this._waterAmount;
  }

  constructor(power) { this._power = power; }
}
```

Protected fields child classes me mil jaate hain (inherit hote hain).

Read-only property

Sirf getter do, setter mat do:

javascript

```
get power() { return this._power; }  
// coffeeMachine.power = 25; -> error
```

`getWaterAmount()` / `setWaterAmount()` jaise functions bhi likh sakte ho. Ye zyada flexible hote hain (kai arguments le sakte hain), get/set syntax chhota hota hai. Dono theek hain.

Private: `#` prefix

Language khud enforce karti hai.

javascript

```
class CoffeeMachine {  
  #waterLimit = 200;  
  
  #checkWater(value) {  
    if (value < 0) return 0;  
    if (value > this.#waterLimit) return this.#waterLimit;  
    return value;  
  }  
}
```

- Bahar se `machine.#waterLimit` karoge to **error**.
- **Child class se bhi access nahi** ho sakta.
- Public aur private same naam se ek saath ho sakte hain (`waterAmount` getter + `#waterAmount` field).
- `this["#name"]` jaisa dynamic access private fields par nahi chalta.

Kyunki child class se bhi access nahi hota, kai baar protected (`␣`) zyada practical lagta hai.

Yaad rakho

- Protected = `␣`, convention, child classes use kar sakti hain.
- Private = `#`, language enforce karti hai, sirf usi class ke andar.
- Bahar ke liye simple public interface rakho, baaki chhupa do.